

MUTUAL FUND ANALYSIS AND PERFORMANCE ANALYTICS USING PYTHON, SQL, AND POWER BI

Project Report

Submitted By

Vishal Hedau

Academic Year

2025–2026

ABSTRACT

The Mutual Fund Analysis and Performance Analytics project aims to analyze mutual fund data using Python, SQL, and Power BI to derive meaningful insights regarding fund performance, risk, and return characteristics. The project involves collecting and cleaning NAV (Net Asset Value) data, storing it in a SQLite database, performing exploratory data analysis (EDA), executing analytical SQL queries, and visualizing results through charts and dashboards.

The developed system helps investors understand fund performance trends, volatility, return distributions, and relationships among different mutual funds. The project demonstrates the application of data analytics, database management, and business intelligence techniques in the financial domain.

Keywords: Mutual Funds, NAV Analysis, Python, SQLite, Power BI, Data Analytics, Performance Evaluation.

CHAPTER 1: INTRODUCTION

1.1 Background

Mutual funds are among the most popular investment instruments because they provide diversification, professional management, and accessibility to investors. Evaluating mutual fund performance requires analyzing historical NAV data, returns, volatility, and risk metrics.

Modern data analytics techniques enable investors and analysts to gain valuable insights from large financial datasets and make informed investment decisions.

1.2 Problem Statement

Investors often find it difficult to compare multiple mutual funds and evaluate their historical performance. Manual analysis is time-consuming and prone to errors.

Therefore, there is a need for an automated system that can:

- Collect mutual fund data
- Clean and process datasets
- Store data efficiently
- Perform performance analysis

- Generate visual insights
- Support decision-making

1.3 Objectives

The objectives of this project are:

1. Collect and organize mutual fund NAV data.
 2. Clean and preprocess financial datasets.
 3. Store data using SQLite.
 4. Perform exploratory data analysis.
 5. Calculate fund performance metrics.
 6. Generate meaningful visualizations.
 7. Build a Power BI dashboard.
 8. Provide insights for investment analysis.
-

CHAPTER 2: SYSTEM REQUIREMENTS

2.1 Software Requirements

- Python 3.x
- Jupyter Notebook
- SQLite
- Power BI Desktop
- Visual Studio Code
- Pandas
- NumPy
- Matplotlib
- Seaborn

2.2 Hardware Requirements

- Intel Core i3/i5 Processor
 - 4 GB RAM or above
 - 10 GB Free Disk Space
 - Windows 10/11
-

CHAPTER 3: DATA COLLECTION AND PREPROCESSING

3.1 Data Collection

Historical NAV data was collected and stored in CSV format.

The dataset contains:

- Date
- Fund Name
- NAV Value

3.2 Data Cleaning

The following preprocessing steps were performed:

- Missing value handling
- Date conversion
- Data type correction
- Duplicate removal
- Daily return calculation
- Infinite value removal

3.3 Cleaned Dataset

The processed dataset was saved as:

clean_nav_data.csv

The dataset contains:

- date
- nav
- fund_name
- daily_return

CHAPTER 4: DATABASE DESIGN

4.1 SQLite Database

A SQLite database was created to store mutual fund information.

Database:

bluestock_mf.db

4.2 Database Tables

nav_history

Stores historical NAV records.

fact_nav

Stores processed NAV information.

fact_performance

Stores calculated performance metrics.

4.3 SQL Queries

Ten analytical SQL queries were developed:

1. Average NAV by fund
2. Record count by fund
3. Maximum NAV
4. Minimum NAV
5. NAV range
6. Average NAV comparison
7. Volatility proxy
8. Date-wise record count
9. Latest NAV records
10. High NAV fund identification

CHAPTER 5: EXPLORATORY DATA ANALYSIS

EDA was performed to understand trends and patterns in mutual fund data.

5.1 NAV Trend Analysis

A line chart was generated to visualize NAV movement over time.

Chart: nav_trend.png

Observation

Most funds demonstrated long-term growth trends.

5.2 Average NAV Analysis

Average NAV values were compared across funds.

Chart: avg_nav.png

Observation

Some funds maintained significantly higher NAV values compared to others.

5.3 Volatility Analysis

Volatility was calculated using the standard deviation of daily returns.

Chart: volatility.png

Observation

Different funds exhibited varying risk levels.

5.4 Return Distribution

Daily return distributions were analyzed.

Chart: return_distribution.png

Observation

Most daily returns were concentrated around zero, indicating relatively stable short-term movements.

5.5 NAV Boxplot

NAV distribution was examined using boxplots.

Chart: nav_boxplot.png

Observation

Certain funds showed wider distributions and larger variability.

5.6 Correlation Matrix

Correlation between fund returns was analyzed.

Chart: correlation_matrix.png

Observation

Several funds displayed strong positive correlations.

CHAPTER 6: PERFORMANCE ANALYTICS

Performance analytics were conducted using total return calculations.

6.1 Total Return Calculation

The following formula was used:

$$\text{Total Return (\%)} = ((\text{Ending NAV} - \text{Starting NAV}) / \text{Starting NAV}) \times 100$$

6.2 Fund Ranking

Funds were ranked according to total returns.

Chart: fund_return_ranking.png

Observation

Top-performing funds generated significantly higher returns than others.

6.3 Volatility Ranking

Funds were ranked based on risk levels.

Chart: volatility_ranking.png

Observation

Higher returns were often associated with higher volatility.

CHAPTER 7: POWER BI DASHBOARD

A Power BI dashboard was developed to provide interactive analysis.

Dashboard File:

bluestock_mf.pbix

Features:

- NAV trends
- Fund comparison
- Performance metrics
- Interactive filtering
- Visual summaries

Benefits:

- Better visualization
 - Faster analysis
 - Investor-friendly interface
-

CHAPTER 8: KEY FINDINGS

1. Most mutual funds demonstrated positive long-term growth.
2. Average NAV values varied significantly across funds.
3. Daily returns were generally centered near zero.
4. Some funds exhibited higher volatility than others.

5. Return correlations were positive among many funds.
 6. Risk and return showed a noticeable relationship.
 7. A few funds consistently outperformed others.
 8. Fund diversification opportunities were identified through correlation analysis.
 9. Historical NAV trends provided valuable performance insights.
 10. Data visualization improved investment decision-making.
-

CHAPTER 9: CONCLUSION

The Mutual Fund Analysis and Performance Analytics project successfully demonstrated how Python, SQL, and Power BI can be integrated to analyze financial datasets.

The project automated data cleaning, database management, exploratory analysis, and performance evaluation. Multiple charts and dashboards were generated to help understand mutual fund behavior and investment patterns.

The developed solution provides an efficient framework for financial analytics and can assist investors in making informed decisions.

CHAPTER 10: FUTURE SCOPE

Future enhancements may include:

1. Live NAV data integration using APIs.
 2. Portfolio optimization techniques.
 3. Machine learning-based return prediction.
 4. Risk-adjusted performance metrics.
 5. Recommendation systems for investors.
 6. Web-based analytics dashboard.
 7. Real-time fund monitoring.
-

REFERENCES

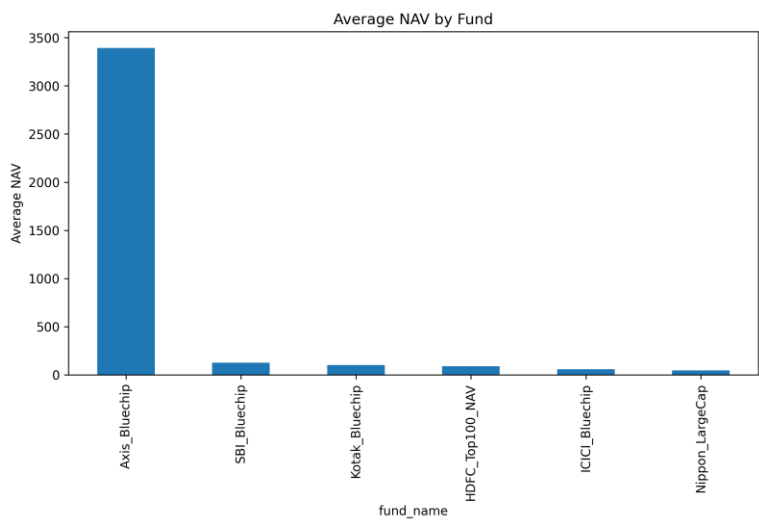
1. Python Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Seaborn Documentation

- 6. SQLite Documentation
- 7. Power BI Documentation
- 8. Mutual Fund Industry Reports

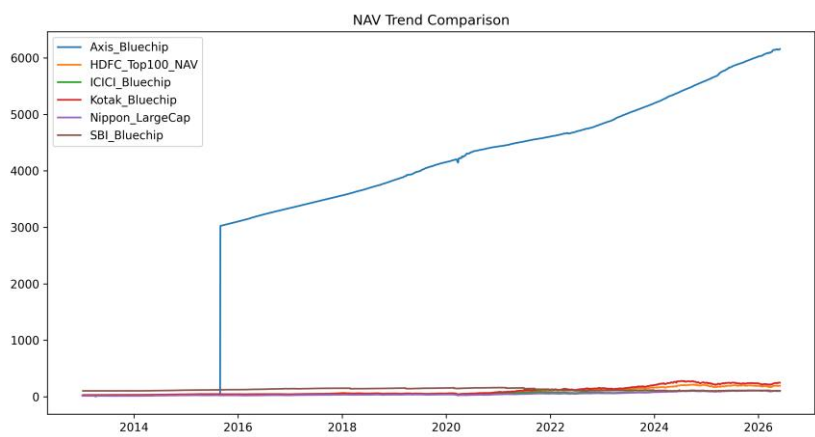
APPENDICES

Generated Charts

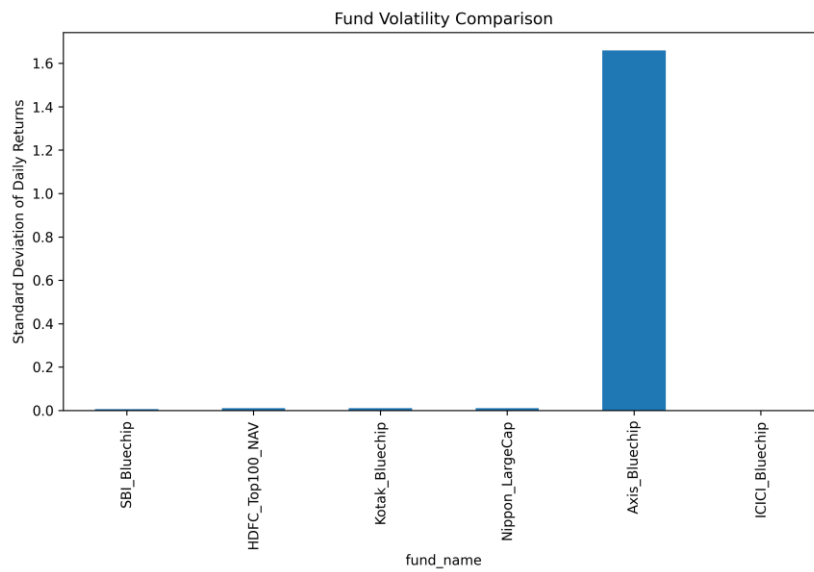
1. avg_nav.png



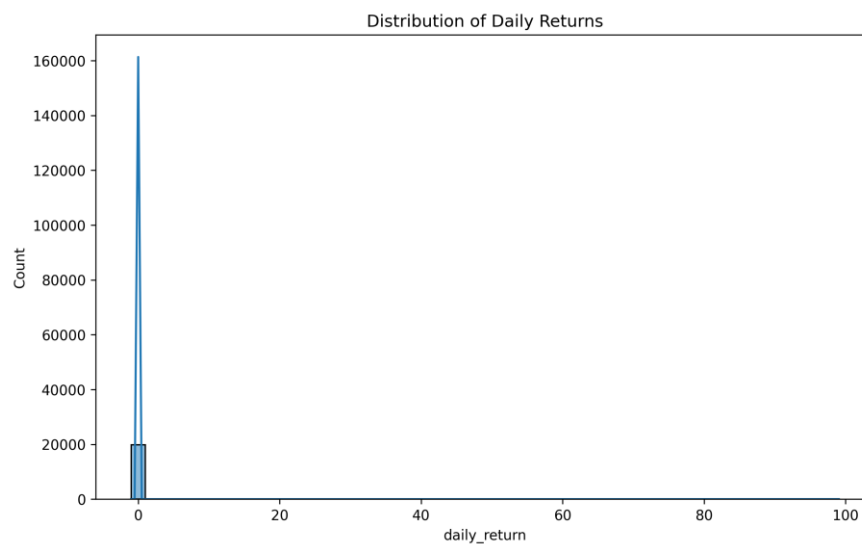
2. nav_trend.png



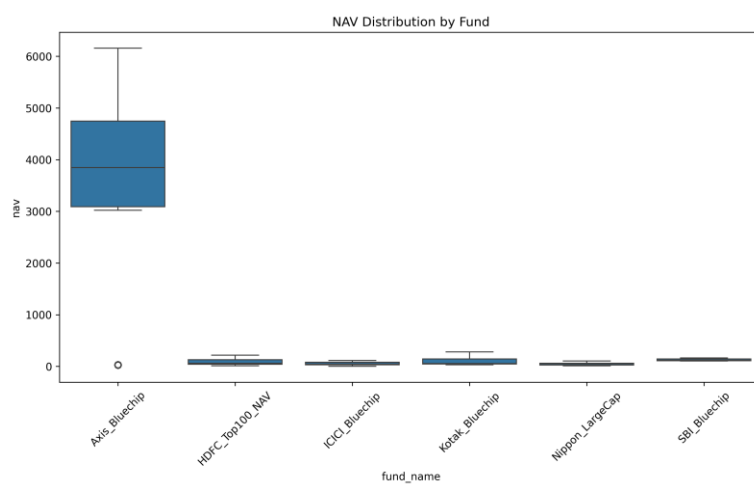
3. volatility.png



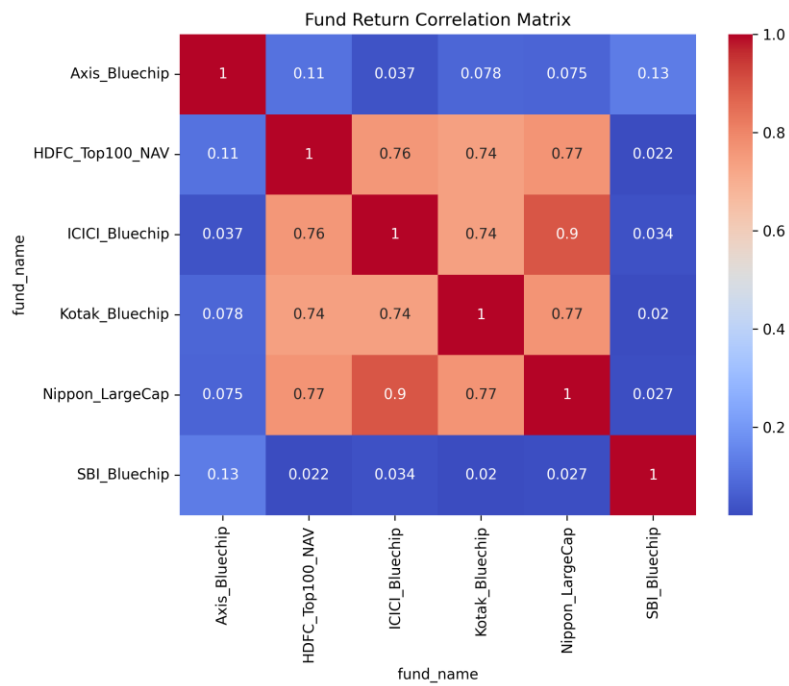
4. return_distribution.png



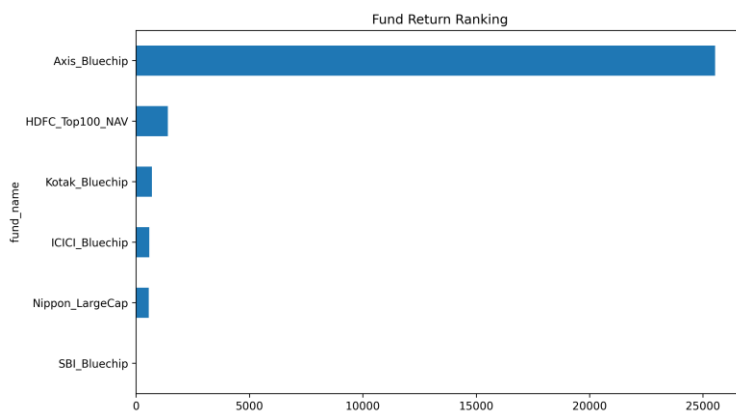
5. nav_boxplot.png



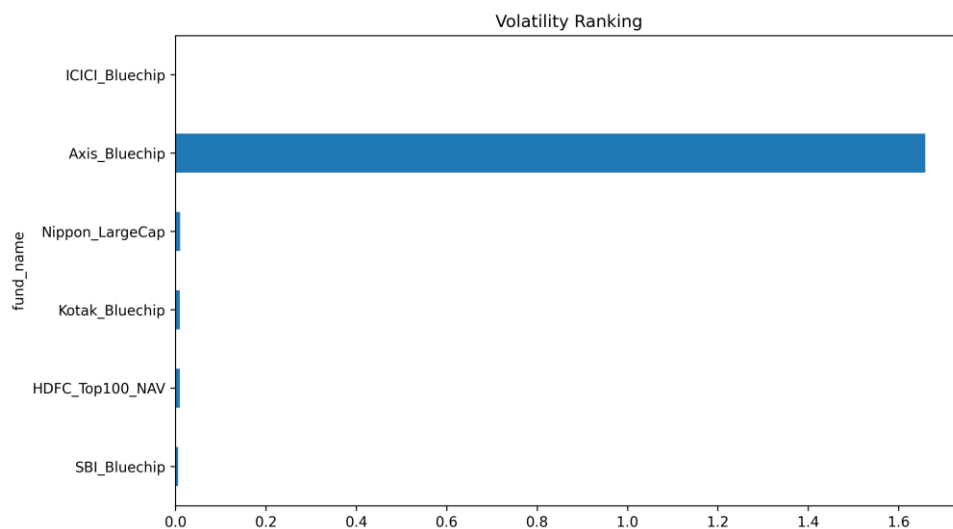
6. correlation_matrix.png



7. fund_return_ranking.png



8. volatility_ranking.png



Project Files

- `clean_nav_data.csv`
- `performance_metrics.csv`
- `bluestock_mf.db`
- `schema.sql`
- `queries.sql`
- `bluestock_mf.pbix`